

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

AI-focused Software Engineer with 9+ years of experience designing scalable systems, intelligent assistants, and ML-backed applications. Proven expertise in Python, LangChain, TensorFlow, and PyTorch for LLM applications; experienced in vector databases (FAISS, Pinecone), prompt engineering, and Retrieval-Augmented Generation (RAG) pipelines. Strong frontend skills in React and Angular, full-stack experience with FastAPI and Node.js, and hands-on with AWS and Docker. Passionate about NLP, chatbot systems, ethical AI, and fine-tuning open-source LLMs for production.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL, C#, Bash
- **AI/ML Tools:** LangChain, TensorFlow, PyTorch, OpenAI, HuggingFace, Scikit-learn, Rasa
- **LLM & NLP:** ChatGPT APIs, Transformers, Sentence Transformers, spaCy, NLTK
- **Vector DBs:** FAISS, Pinecone, Weaviate
- **Databases:** PostgreSQL, MongoDB, Cosmos DB
- **Frameworks:** FastAPI, Next.js, React, Angular, Node.js, .NET Core
- **DevOps & Cloud:** AWS, GCP, Azure, Docker, Kubernetes, GitHub Actions, CI/CD
- **Architectures:** Microservices, REST/gRPC APIs, RAG pipelines
- **Other:** Redis, Kafka, WebSockets, Figma (UX), OpenTelemetry

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Architected AI-powered chatbot solutions using LangChain and OpenAI for enterprise support, integrating Pinecone for context retrieval and embedding management.
- Led migration of legacy systems to microservices using FastAPI, improving scalability and integrating gRPC-based communication for AI agents.
- Developed internal RAG framework combining HuggingFace models and Weaviate to power document Q&A and multilingual virtual assistant flows.
- Designed and built vector-based semantic search for over 10K product manuals using FAISS with hybrid search fallback on PostgreSQL.
- Integrated LLMs with business processes by creating secure prompt templates and orchestrating dynamic flows with LangChain agents.

- Delivered secure, scalable frontend interfaces using React and Next.js for client-facing AI dashboards and feedback capture tools.
- Deployed NLP services via AWS Lambda and ECS, with CI/CD pipelines on GitHub Actions and monitoring via CloudWatch and Prometheus.
- Mentored junior engineers in LLM fine-tuning, Python best practices, and modular component design for both backend and frontend.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Built React-based UI for a smart advertising engine, integrating real-time personalization powered by ML pipelines on TensorFlow.
- Implemented robust REST APIs for campaign orchestration and user behavior analytics using FastAPI, PostgreSQL, and Redis.
- Created pipelines for keyword extraction and text sentiment analysis using spaCy and Vader for campaign optimization.
- Collaborated with data scientists to preprocess and vectorize large corpora, integrating FAISS search with real-time filters.
- Containerized all backend components with Docker and maintained Azure-based deployment workflows with GitLab CI/CD.
- Ensured GDPR compliance across ML models with explainability layers and opted-in data pipelines for user-level predictions.
- Prototyped a voice assistant using Google's Speech-to-Text API and OpenAI's GPT to process call center FAQs.
- Provided detailed documentation and architectural diagrams using PlantUML and Mermaid for smooth team handoffs.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Designed scalable backend services in Node.js to support custom-built web crawlers and NLP-powered data tagging systems.
- Engineered multimodal interfaces combining images and text metadata, storing vectors in Pinecone for cross-modal queries.
- Collaborated on early prototypes of voice-interactive dashboards with embedded AI reasoning from pre-trained transformer models.
- Introduced CI/CD with Jenkins and Docker Compose, significantly reducing environment mismatch issues across teams.
- Conducted A/B testing pipelines in the frontend using Angular and React to validate user engagement for recommendation widgets.
- Delivered data collection apps with offline syncing using PouchDB and CouchDB for rural deployment scenarios.
- Developed user permission systems using JWT-based access and OAuth2 integration with Google and Microsoft accounts.
- Monitored distributed systems using ELK Stack, writing custom parsers for structured logs emitted by AI inference services.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Developed AI data pipelines for chatbot sentiment tagging and integrated with existing CRM infrastructure via REST APIs.
- Worked on Kafka-enabled data processing services to log and evaluate real-time chat events for escalation prediction.
- Applied Protobuf for schema control and high-efficiency message transmission between analytics services.
- Automated test suites for REST services with Postman and PyTest, ensuring stability across refactor cycles.
- Authored internal tooling for prompt version testing and dialogue flow simulation using Flask and React.
- Supported rollout of NLP services in QA and staging via Docker Swarm and Ansible playbooks.
- Built small-scale dashboard using Angular to monitor user interaction rates and classify unresolved chatbot intents.
- Ran performance audits on deployed services using Apache Bench and New Relic instrumentation.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Assisted in training custom text classification models using scikit-learn and curated datasets for niche legal use cases.
- Developed internal dashboards to visualize Kafka topic metrics and consumer lags using open-source tooling.
- Contributed to proof-of-concept projects integrating Kafka with IoT sensor data for stream-based analytics.
- Maintained a centralized repository of reusable Python data wrangling scripts for model prep automation.
- Helped with React-based internal tool development to track annotation completeness and user feedback.
- Performed comparative experiments with TF-IDF and BERT-based embeddings for question answering pipelines.
- Supported model evaluation by implementing ROC, precision-recall, and confusion matrix visualizers.
- Wrote technical documentation and experiment logs to guide team members in re-running NLP pipelines.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ **TensorFlow Developer Certificate – DeepLearning.AI**
 - ❖ **LangChain for LLM Application Development – DeepLearningCamp**
 - ❖ **AWS Certified Cloud Practitioner – Amazon Web Services**
 - ❖ **Prompt Engineering for LLMs – DeepLearning.ai / OpenAI**
 - ❖ **Docker & Kubernetes for Developers – Udemy**
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PROJECTS

- ✓ **LegalBot – AI Legal Assistant**
Built a multilingual chatbot using LangChain, GPT-4, FAISS, and Streamlit to answer legal queries for SMEs, integrating PDF processing and OpenAI function calling.
- ✓ **VectorDoc Search Engine**
Developed a document understanding platform using HuggingFace transformers and Pinecone, enabling semantic document search and dynamic RAG responses for internal reports.
- ✓ **Voice-to-Action Assistant**
Prototyped a voice-activated assistant using OpenAI Whisper, Flask, and FastAPI that handled meeting summarization and contextual task generation with GPT integration.